CSM
Fall 2026

EE 64
Worksheet 4

1. Beginning Circuit Analysis

Meta: The goal of this worksheet is to **familiarize students with NVA and passive sign convention and to calculate node voltages based on Ohm's Law, passive sign convention, KCL, and KVL. The point is not to actually solve for the node voltages.**

Some items to remind students of if they are struggling initially:

- Procedure for NVA includes labeling the nodes, labeling currents according to passive sign convention (+ to -), and labeling voltages across circuit elements
- KCL: current flowing into a node = flowing out of a node
- KVL: sum of voltages across the elements connected in a loop is equal to zero.

Before starting, make sure to do a mini-lecture or review on:

- The IV relationship between a voltage source, current source, resistor, capacitor, open circuit, and write
- Units and symbols for voltage, current, and resistance
- Ohm's Law
- Review of KCL and KVL

(a) Follow the directions for this subpart to label the circuit according to passive sign convention and NVA.

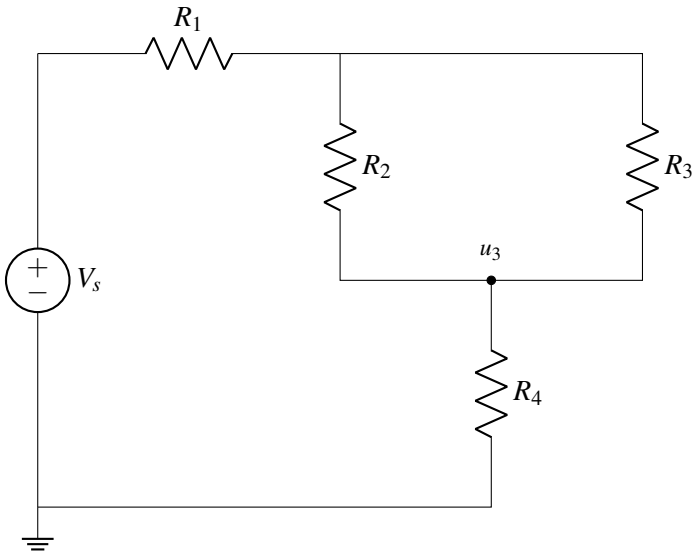
i. Label each node of the of the following circuit with the following notation of u_i.

ii. Label each side of the circuit components either positive (+) or negative (-).

iii. Represent currents as arrows and draw the currents that flow through each circuit component, denoted as i_i. *Use your labeling from the previous part. Currents flow according to passive sign convention.*

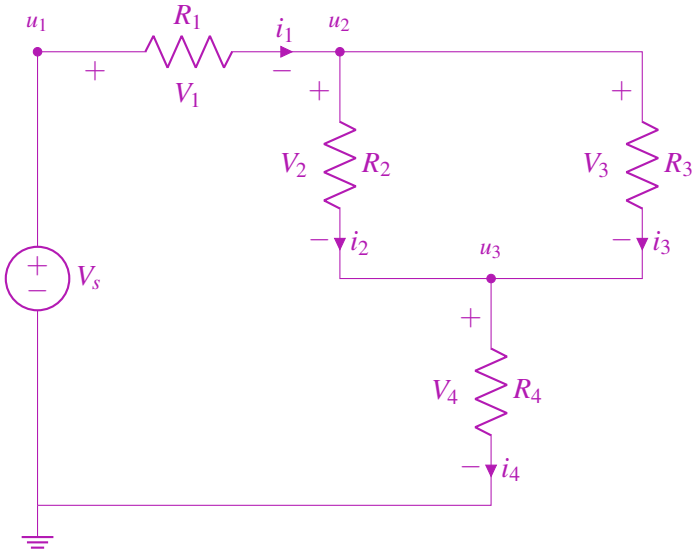
iv. Label the voltages across each circuit, denoted as V_i.

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Meta: Emphasize that they just need to keep their labeling **consistent**. They will get the same answer later anyways.

Solution: Please note that the ground node was omitted during the process of labeling. This is because rather than maybe writing a voltage across a circuit element as $u_i - u_{ground}$, it can be simplified to u_i because u_{ground} is 0V. By this logic, node u_1 is equal to V_s.
It's also okay if you labeled the circuit differently from this! As long as you adhere to passive sign convention and stay consistent with your labeling, you should arrive at the same answer for parts (b) and (c). Your final circuit should resemble something like the labeled circuit below with currents, voltages, and nodes labeled, although its okay if it is not exactly like it.



(b) Now that we've correctly labeled our circuit, lets begin the circuit analysis section!

Use KCL to determine the value of u_3 node in terms of any of the following: *other node voltages*, R_1, R_2, R_3, R_4 and V_s (not currents though!!). **Label circuit potential differences and currents.** *Hint: Write out your KCL or KVL equations according to NVA and passive sign convention. How can you*

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manipulate these equations into solving for u_3?

Solution: For KCL equations, we write it as current(s) flowing into a node being equal to the current(s) flowing out of the node.
KCL at u_2:

$$i_1 = i_2 + i_3 \tag{1}$$

KCL at u_3:

$$i_2 + i_3 = i_4 \tag{2}$$

Ohm's Law states that:

$$V = IR \tag{3}$$

which V is voltage (V), I is current (A), and R is resistance (ohms). This can also be rearranged to solve for current so that:

$$I = \frac{V}{R} \tag{4}$$

Rewrite equation (2), the second KCL equation, since it includes i_4. This is useful because i_4 can be simplified using Ohm's Law to get u_3. $i_4 = \frac{u_3}{R_4}$, which includes the desired node u_3:

$$\frac{u_3}{R_4} = \frac{u_2 - u_3}{R_2} + \frac{u_2 - u_3}{R_3} \tag{5}$$

Simplifying:

$$u_3 = (u_2 - u_3)\left(\frac{R_4}{R_2} + \frac{R_4}{R_3}\right) \tag{6}$$

$$u_3 + u_3\left(\frac{R_4}{R_2} + \frac{R_4}{R_3}\right) = u_2\left(\frac{R_4}{R_2} + \frac{R_4}{R_3}\right)$$

$$u_3 = u_2\left(\frac{R_4/R_2 + R_4/R_3}{1 + \frac{R_4}{R_2} + \frac{R_4}{R_3}}\right)$$

Depending on how you rearranged your KCL equations you may have a different equation. However, if your signs and methodology is correct, then if given real values, you would get the **same answer**.

(c) Let's analyze this circuit utilizing KVL now! As a reminder, you can write a KVL equation for a closed loop.

Use KVL to determine the value of u_3 node in terms of *currents (not other node voltages),*, R_1, R_2, R_3, R_4 and V_s. Please do not include node voltages in your final answer.

Meta: Emphasize that answers may be different from each other, but **as long as they follow how they labeled their circuit and don't mess up on the signs / Ohm's Law, they have a valid answer.**

Solution: For KVL equations, we write the equations as a sum of voltages across the elements connected in a loop as being equal to zero. Going from a the (+) terminal to the (-) terminal means that you are going down in potential, so place a negative sign in front of the voltage for the circuit element.

KVL equations:

$$V_s - V_1 - V_2 - V_4 = 0 \tag{7}$$

$$V_s - V_1 - V_3 - V_4 = 0 \tag{8}$$

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We should recognize that the voltage at node u_3 is equal to the the voltage across the resistor V_4 since $V_4 = u_3 - 0$. Therefore, we can rearrange for equation (7) to solve for node u_3.

$$u_3 = V_s - V_1 - V_3 \tag{9}$$

Utilizing equation (3), Ohm's Law, we can rewrite the solution as:

$$u_3 = V_s - i_1 R_1 - i_3 R_3 \tag{10}$$

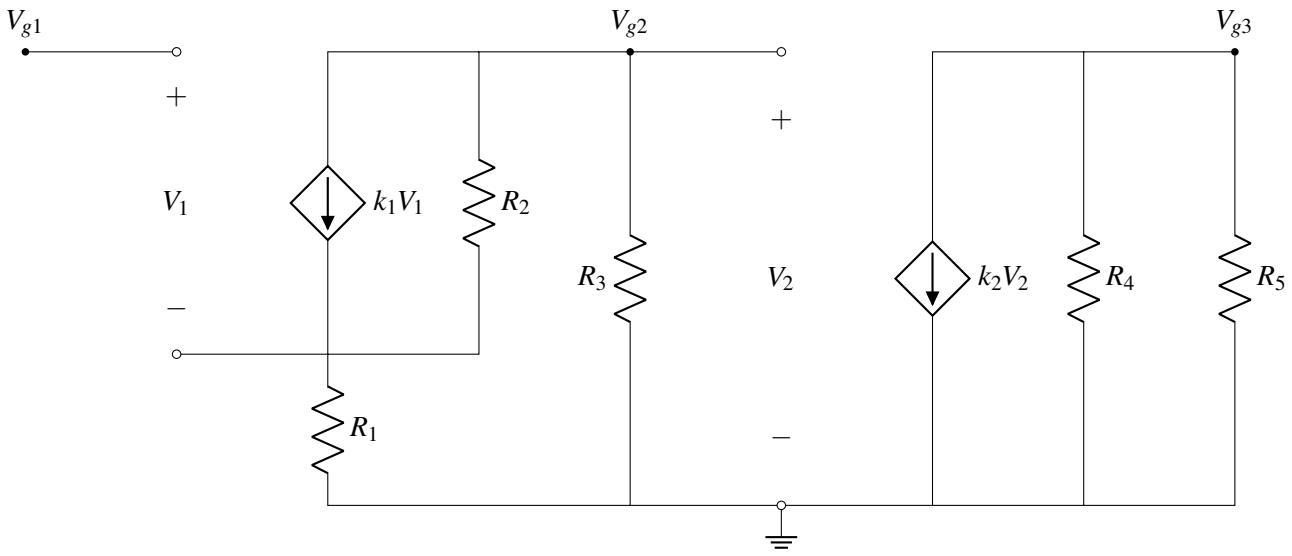
Depending on how you rearranged your KVL equations you may have a different equation. However, if your signs and methodology is correct, then if given real values, you would get the **same answer**.

Depending on the circuit and what the question is asking you to calculate for, it may be easier to use KCL or KVL. However, if actually solving to get a numerical value, following NVA and passive sign convention consistently will get you to the **same answer** anyways. One process may have more steps than the other.

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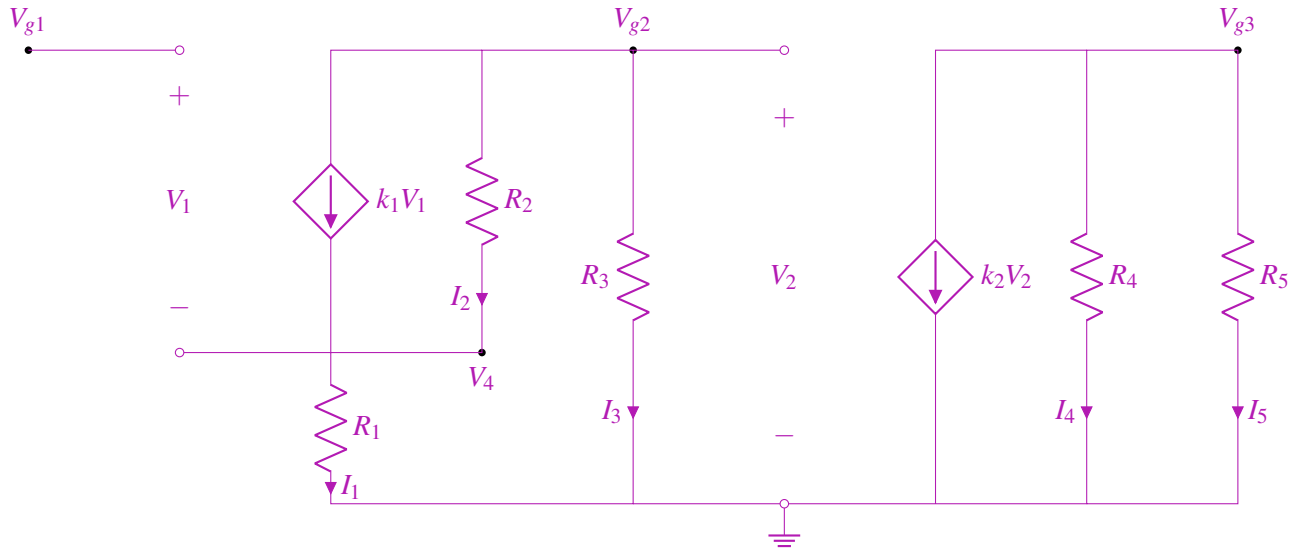
2. Kirchhoff's Laws with Dependent Sources

Consider the following circuit:



(a) Solve for $\frac{V_{g2}}{V_{g1}}$ using nodal analysis.

Solution: Label the current through each resistor, and denote the node voltage below resistor 2 as V_4.



KCL:

$$k_1V_1 + I_2 + I_3 = 0$$

$$k_1V_1 + I_2 - I_1 = 0$$

The first equation gives us:

$$k_1(V_{g1} - V_4) + \frac{V_{g2} - V_4}{R_2} + \frac{V_{g2}}{R_3} = 0$$

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$$V_4 = \frac{k_1 V_{g1} + V_{g2}(\frac{1}{R_2} + \frac{1}{R_3})}{k_1 + \frac{1}{R_2}}$$

The second equation gives us:

$$k_1(V_{g1} - V_4) + \frac{V_{g2} - V_3}{R_2} - \frac{V_4}{R_1} = 0$$

$$k_1 V_{g1} + \frac{V_{g2}}{R_2} = V_4(k_1 + \frac{1}{R_2} + \frac{1}{R_1})$$

Plug in the value of V_4,

$$k_1 V_{g1} + \frac{V_{g2}}{R_2} = \left(\frac{k_1 V_{g1} + V_{g2}(\frac{1}{R_2} + \frac{1}{R_3})}{k_1 + \frac{1}{R_2}}\right)(k_1 + \frac{1}{R_2} + \frac{1}{R_1})$$

$$V_{g1}\left(\frac{-\frac{k_1}{R_1}}{k_1 + \frac{1}{R_2}}\right) = V_{g2}\left(\frac{\frac{1}{R_1 R_2} + \frac{1}{R_3}(k_1 + \frac{1}{R_1} + \frac{1}{R_2})}{k_1 + \frac{1}{R_2}}\right)$$

$$\frac{V_{g2}}{V_{g1}} = \frac{-k_1 R_2 R_3}{k_1 R_1 R_2 + R_1 + R_2 + R_3}$$

(b) Solve for $\frac{V_{g3}}{V_{g2}}$ using nodal analysis.

Solution: KCL:

$$k_2 V_{g2} + I_4 + I_5 = 0$$

$$k_2 V_{g2} + \frac{V_{g3}}{R_4} + \frac{V_{g3}}{R_5} = 0$$

$$k_2 V_{g2} = -V_{g3}(\frac{1}{R_4} + \frac{1}{R_5})$$

$$\frac{V_{g3}}{V_{g2}} = \frac{-k_2 R_4 R_5}{R_4 + R_5}$$

(c) Solve for $\frac{V_{g3}}{V_{g1}}$ using the results above.

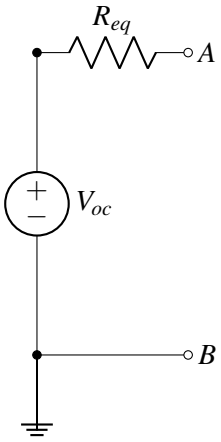
Solution:

$$\frac{V_{g3}}{V_{g1}} = \frac{V_{g3}}{V_{g2}} \cdot \frac{V_{g2}}{V_{g1}}$$

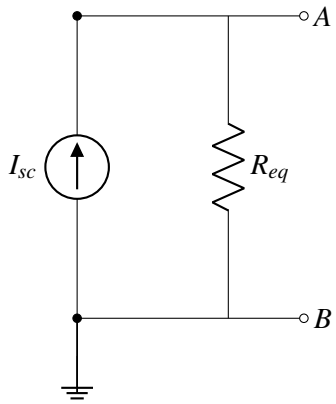
$$\frac{V_{g3}}{V_{g1}} = \frac{k_1 k_2 R_2 R_3 R_4 R_5}{(k_1 R_1 R_2 + R_1 + R_2 + R_3)(R_4 + R_5)}$$

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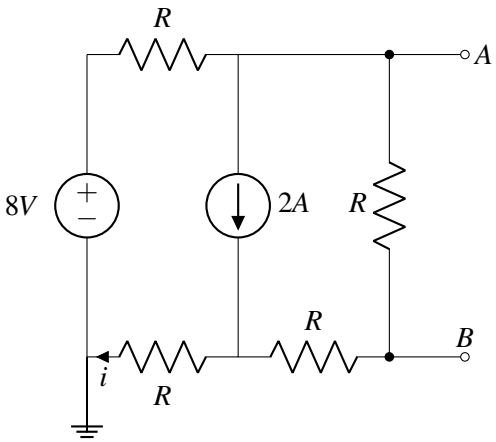
3. Thevenin Norton In this question, you will explore Thevenin Norton equivalence. This is a way of represent the input/output characteristics of a complicated circuit by modeling it as a simple voltage/current source and resistor. Suppose you are given a circuit and 2 terminals A and B. The Thevenin equivalent circuit with respect to those terminals is a circuit of the following form:



where V_{OC} and R_{eq} are set so that the open circuit (OC) voltage between A and B, as well as the short circuit (SC) current through A and B, are the same as in the original circuit. And the Norton equivalent circuit is a circuit of the following form, with the same properties:



(a) Consider the following circuit. Find the equivalent resistance across the A and B terminal.



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Meta: Ensure that the students understand how to convert back and forth between the two forms and where it is appropriate to use one over other. They should also be comfortable with finding resistances when voltage/current sources are involved.

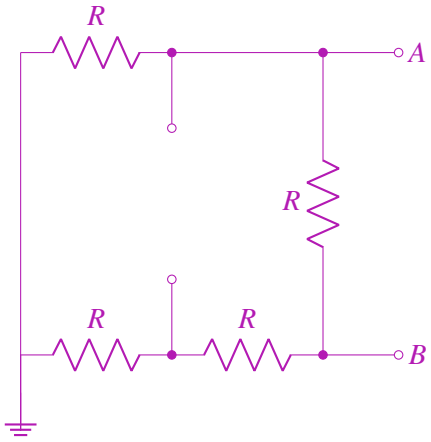
Also, probe their understanding about what happens if dependent sources are involved.

Meta: **Be careful** when using the equation $R_{th} = \frac{V_{oc}}{I_{sc}}$ because this fails under certain circumtances (see last semester's MT2 Thevenin equivalence question). It's fine to teach them this shortcut to use instead of V_{test} and I_{test}, but make sure to emphasize that $V_{\text{test}}/\ I_{\text{test}}$ never fails.

Solution: The first thing we want to do while solving for the equivalences is null out all our sources. **This is only possible if all the sources in the circuit are independent sources**.

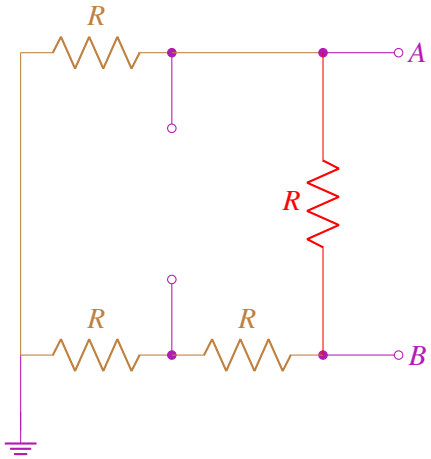
Recall that a null-ed voltage source is a short (because it adds no voltage, it is effectively a wire) and a null-ed out current source is an open circuit (because no current can flow).

The resulting circuit would be:



Now that we have drawn our circuit, we need to trace all possible paths between A and B and calculate our equivalent resistance.

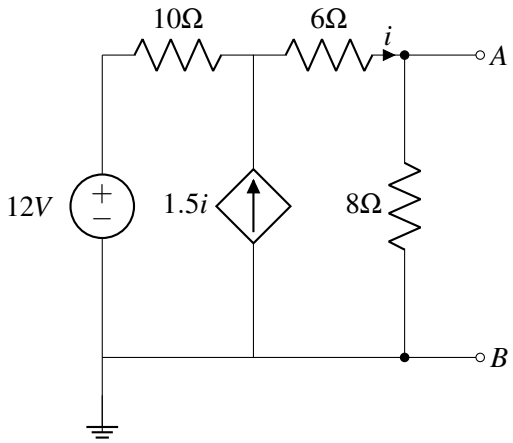
We see we have the two paths colored below:



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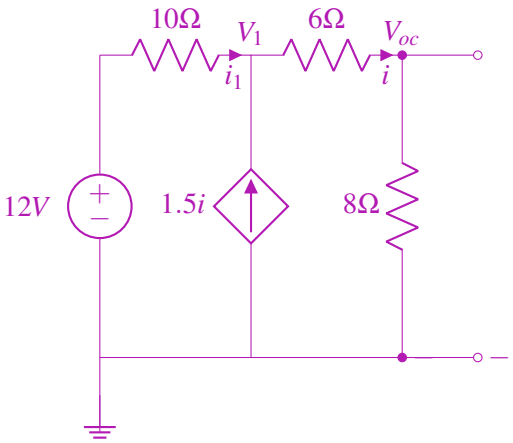
And so we can calculate $R_{eq} = R \parallel (R+R+R) = \frac{1}{\frac{1}{R}+\frac{1}{3R}} = \frac{1}{\frac{4}{3R}} = \frac{3}{4}R$

(b) Consider the following circuit. Find the Thevenin and Norton equivalent circuits.



Solution:

Let us define some additional currents and voltages so that we can perform nodal analysis:



i) First, we find the Thevenin voltage, which is the open circuit voltage labeled V_{oc} across the 8Ω resistor.

Let us perform nodal analysis at V_2;
KCL gives us:

$$i_1 + 1.5i = i$$

From Ohm's Law, we have

$$i_1 = \frac{12V - V_1}{10\Omega}$$

$$i = \frac{V_1 - V_{oc}}{6\Omega} = \frac{V_{oc}}{8\Omega}$$

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Note that the current across the 6Ω resistor is the same as the current across the 8Ω resistor because resistors in series share a current.
First, let us plug in our current values into the KCL equation:

$$\frac{12V - V_1}{10\Omega} + 1.5\frac{V_{oc}}{8\Omega} = \frac{V_{oc}}{8\Omega} \tag{1}$$

Then let's solve for V_1 in terms of V_{oc}:

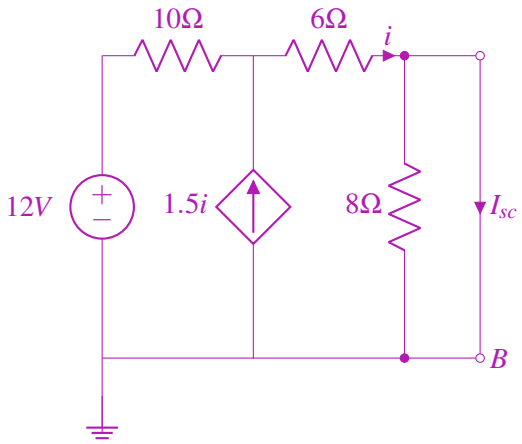
$$V_1 - V_{oc} = \frac{6V_{oc}}{8\Omega}$$

$$V_1 = \frac{6V_{oc}}{8\Omega} + V_{oc} = \frac{14V_{oc}}{8\Omega}$$

Then, substitute V_1 into equation (1):

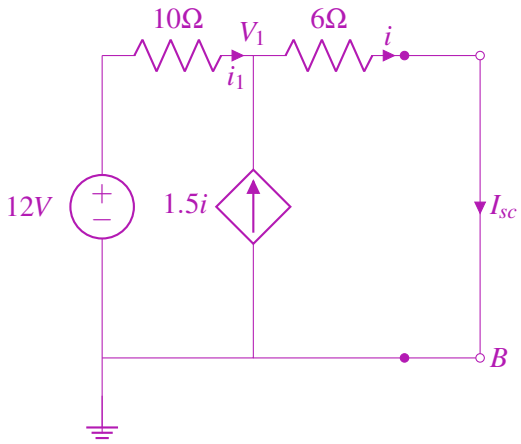
$$\frac{12V - \frac{14V_{oc}}{8\Omega}}{10\Omega} + 1.5\frac{V_{oc}}{8\Omega} = \frac{V_{oc}}{8\Omega}$$

Solving for V_{oc}, we get $V_{oc} = \frac{96}{9}V$, or approximately $10.67V$.
ii) Next, we want to find I_{sc} at the terminals. We want to create a short at the terminals.



Then, the resistor is no longer on the path of the current. Let us also define some additional currents and voltages so that we can perform nodal analysis:

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Now, using KCL at V_1, we get the equation:

$$i_1 + 1.5i = I_{sc}$$

Using Ohm's law, we can plug in the values for i_1:

$$\frac{12V - V_1}{10} = i_1$$

Also, note that $i = I_{sc}$
Solve this equation for V_1.

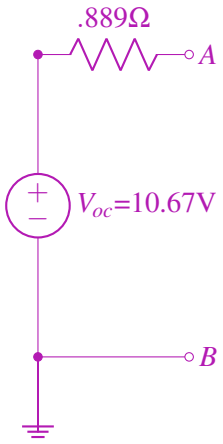
$$\frac{12V - V_1}{10} + 1.5\frac{V_1}{6} = \frac{V_1}{6}$$

$$V_1 = 72V$$

Then, solve for I_{sc}: $I_{sc} = i = \frac{V_1}{6}$
$I_{sc} = \frac{72}{6} = 12A$
Now, we can solve for R_{eq}: $R_{eq} = \frac{V_oc}{I_sc} = \frac{96/9}{12} = 0.889$

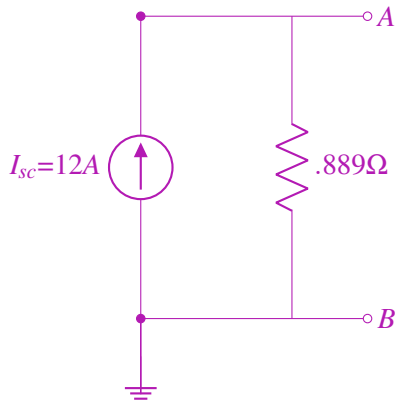
(c) Now, draw the Thevenin and Norton Equivalent Circuit

Solution: Thevenin Equivalent Circuit:



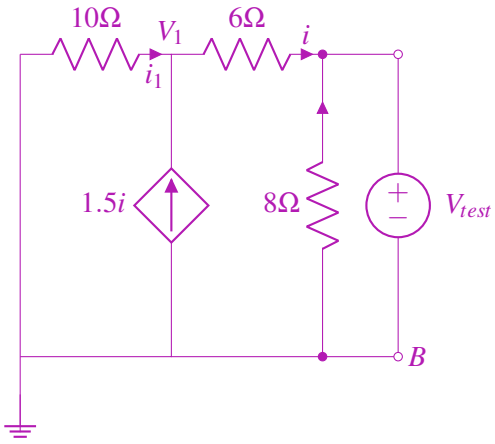
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Norton Equivalent Circuit:



(d) You can find R_{eq} by taking the ratio of $\frac{V_{OC}}{I_{SC}}$ from the previous part. But in this part, we will find R_{eq} directly by using test voltages and currents. The idea is to zero out all independent sources in the circuit (not dependent sources though). Then, we attach a test voltage V_{test} between A and B and calculate the value of I_{test}, which is the current through your test voltage. R_{eq} will be equal to the ratio $\frac{V_{test}}{I_{test}}$. Find the value of R_{eq} by this method, and compare it with the value of $\frac{V_{OC}}{I_{SC}}$

Solution:



Applying KCL at the V_1 node:

$$\frac{0-V_1}{10}+1.5i=i$$
$$\frac{V_1-V_{test}}{6}=i$$

Solving, we get

$$\frac{V_1}{12}-\frac{V_1}{10}-\frac{V_{test}}{12}=0$$
$$\implies \frac{-V_1}{60}=\frac{V_{test}}{12}$$
$$\implies V_1=-5V_{test}$$

Now, we can solve for the current between A and B.

$$i=\frac{V_1-V_{test}}{6}=\frac{-5V_{test}-V_{test}}{6}=-V_{test}$$

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$$i_{BA} = \frac{V_{test} - 0}{8} - i = \frac{V_{test}}{8} + V_{test} = \frac{9}{8} V_{test}$$

So, $R_{eq} = \frac{V_{test}}{i_{BA}} = \frac{8}{9} = 0.889$, which is the same as what we got before.

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